

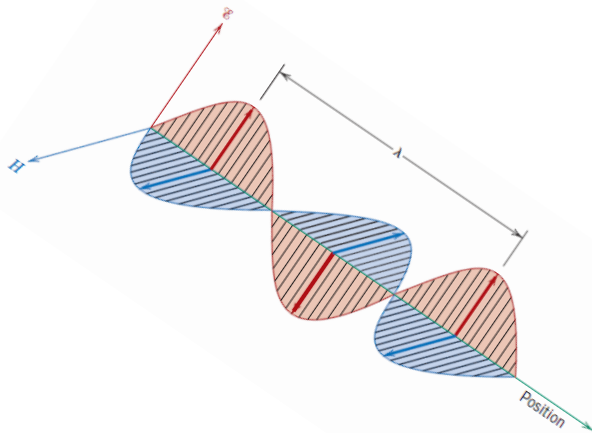
Chapter 10: Optical Properties

ISSUES TO ADDRESS...

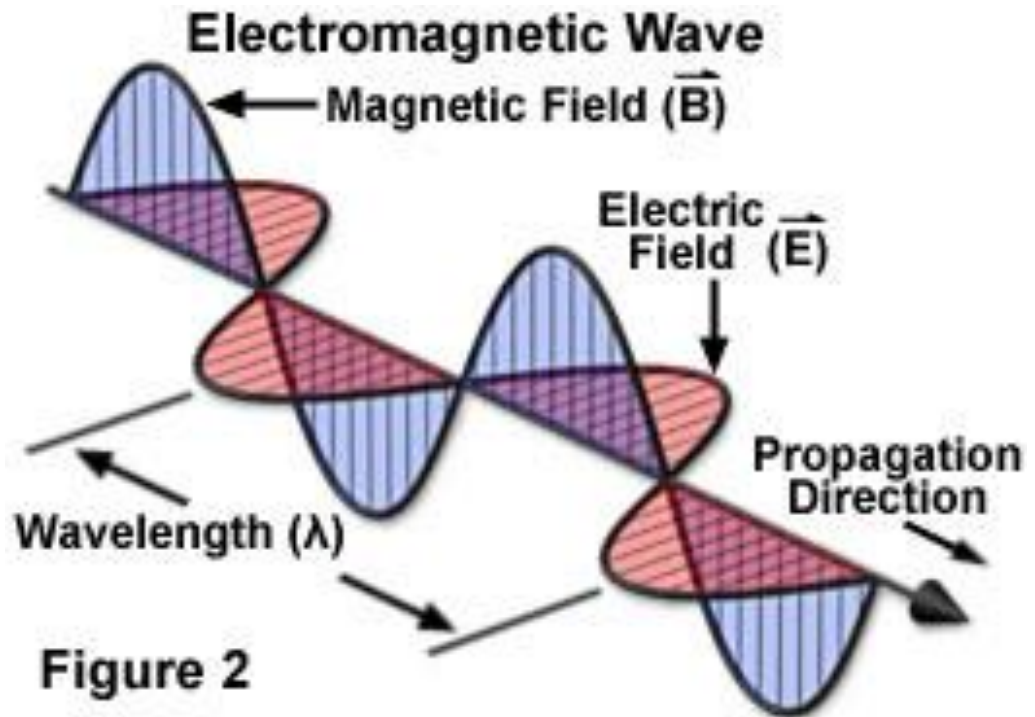
- What phenomena occur when light is shined on a material?
- What determines the characteristic colors of materials?
- Why are some materials transparent and others are translucent or opaque?
- How does a laser operate?

Electromagnetic radiation

- In the classical sense, electromagnetic radiation is considered to be wave-like, consisting of electric and magnetic field components that are perpendicular to each other and also to the direction of propagation.

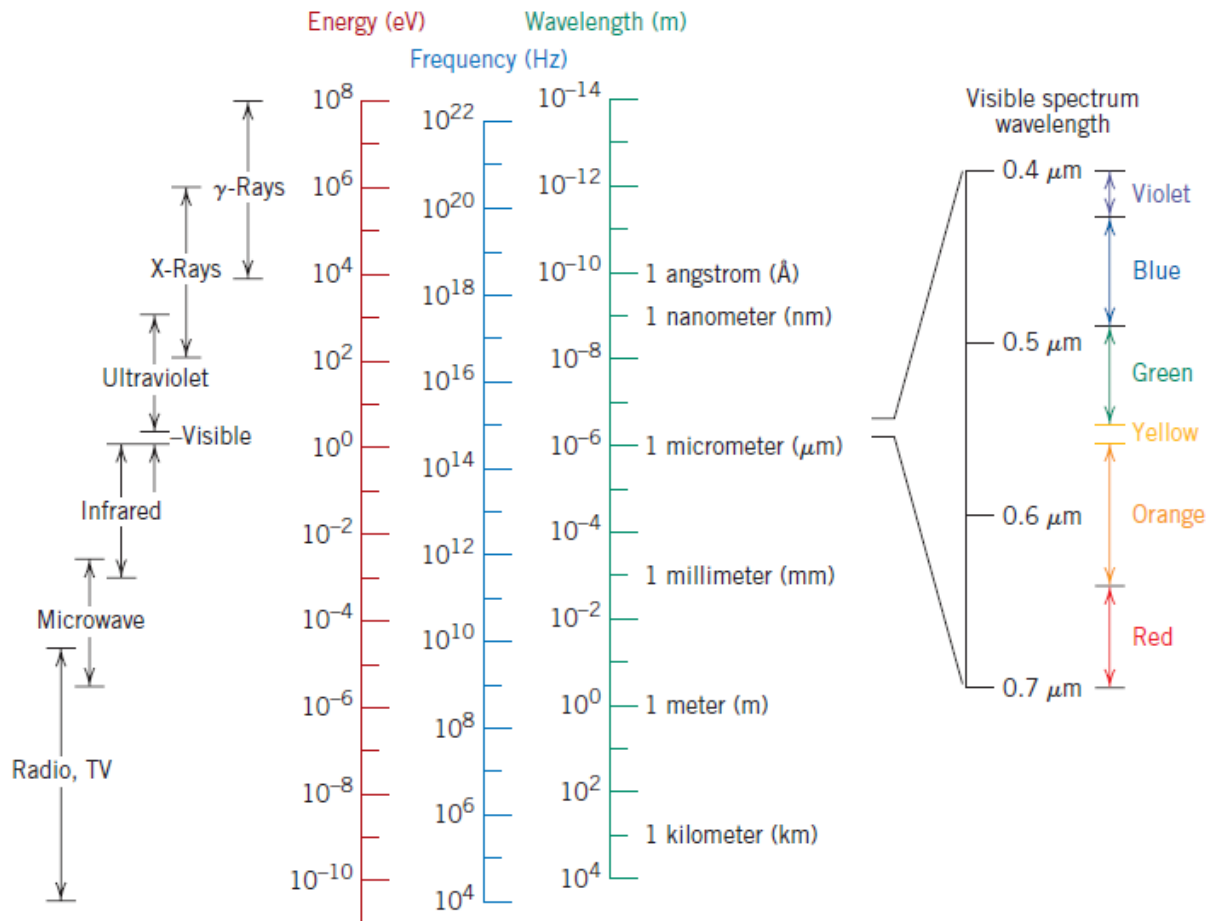


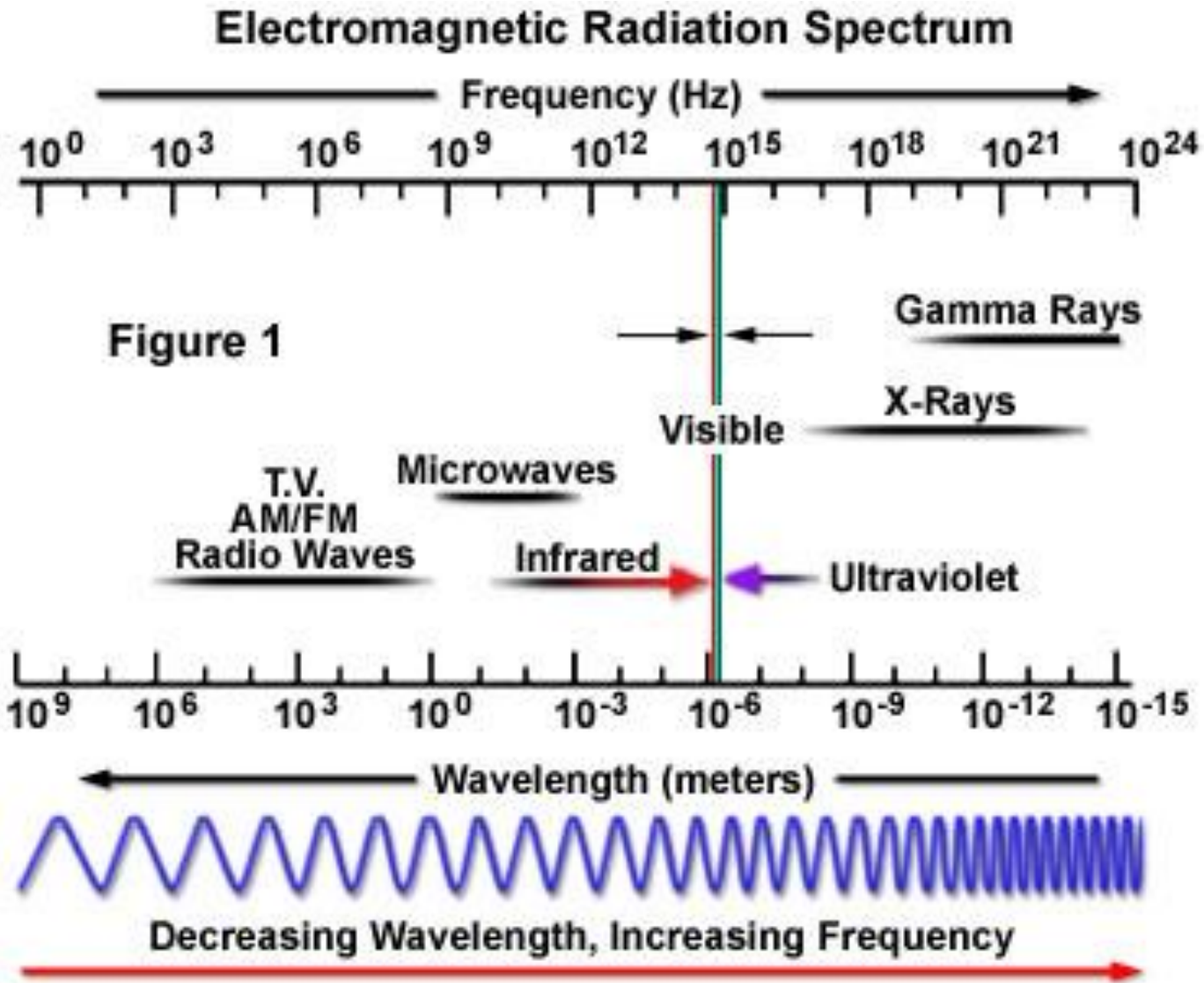
<https://www.olympus-lifescience.com/en/microscope-resource/primer/lightandcolor/electromagneticintro/>



Optical Properties

- Light, heat, radar, radio waves, and x-rays are all forms of electromagnetic radiation.
- Each is characterized primarily by a specific range of wavelengths, and also according to the technique by which it is generated.





Optical Properties

- All electromagnetic radiation traverses a vacuum at the same velocity, that of light—namely, $3 \times 10^8 \text{ m/s}$.
- This velocity, c , is related to the “**electric permittivity**” of a vacuum and the “**magnetic permeability**” of a vacuum through

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

- There is an association between the electromagnetic constant c and these electrical and magnetic constants

Optical Properties

Light has both particulate and wavelike characteristics

– Photon - a quantum unit of light

$$E = h\nu = \frac{hc}{\lambda}$$

E = energy of a photon

λ = wavelength of radiation

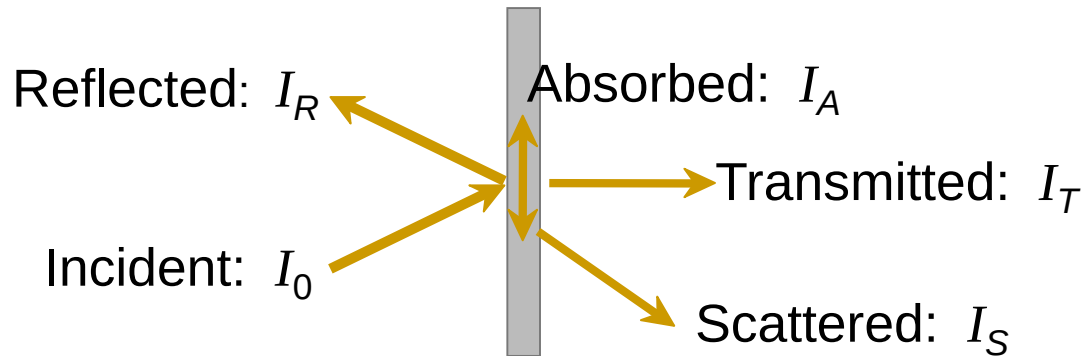
ν = frequency of radiation

h = Planck's constant ($6.62 \times 10^{-34} \text{ J} \cdot \text{s}$)

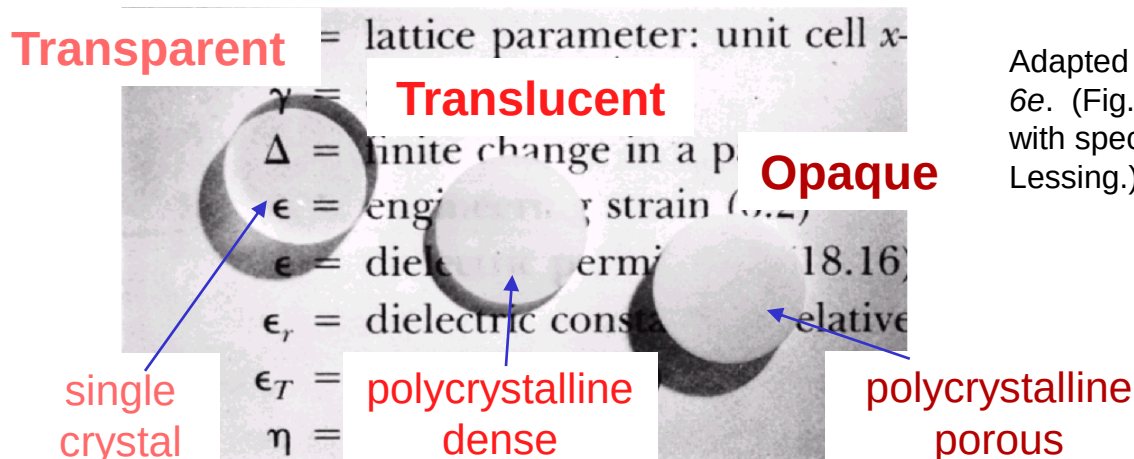
c = speed of light in a vacuum ($3.00 \times 10^8 \text{ m/s}$)

Light Interactions with Solids

- Incident light is reflected, absorbed, scattered, and/or transmitted: $I_0 = I_T + I_A + I_R + I_S$



- Optical classification of materials:



Adapted from Fig. 21.10, *Callister* 6e. (Fig. 21.10 is by J. Telford, with specimen preparation by P.A. Lessing.)

Atomic and Electronic Interactions

Two of the most important interactions are

- Electronic polarization
- Electron energy transitions

-Electronic polarization:

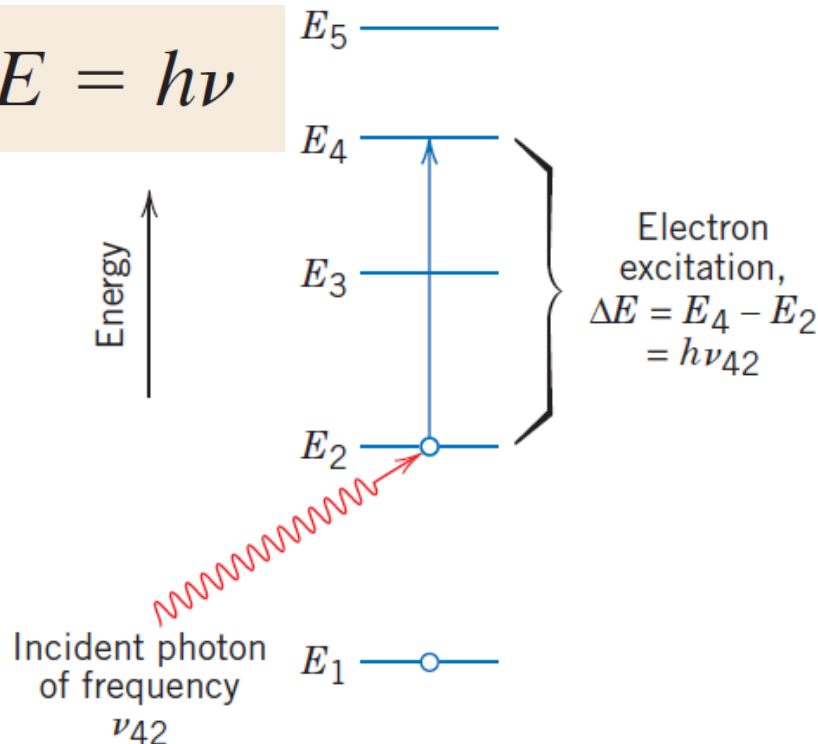
For the visible range of frequencies, electric field component of electromagnetic wave (light) **interacts with the electron cloud** surrounding each atom within its path in such a way as to “**induce electronic polarization**”, or to shift the electron cloud relative to the nucleus of the atom with each change in direction of electric field component.

- Two consequences of this polarization are:
 - (1) some of the radiation energy may be absorbed, and
 - (2) light waves are retarded in velocity as they pass through the medium.

-Electron transitions:

- The **absorption** and **emission** of electromagnetic radiation may involve electron transitions from one energy state to another.
- Consider an isolated atom, the electron energy diagram. An electron may be excited from an occupied state at energy E_2 to a vacant and higher-lying one, denoted E_4 by the **absorption** of a photon of energy.
- The change in energy experienced by the electron, depends on the radiation frequency:

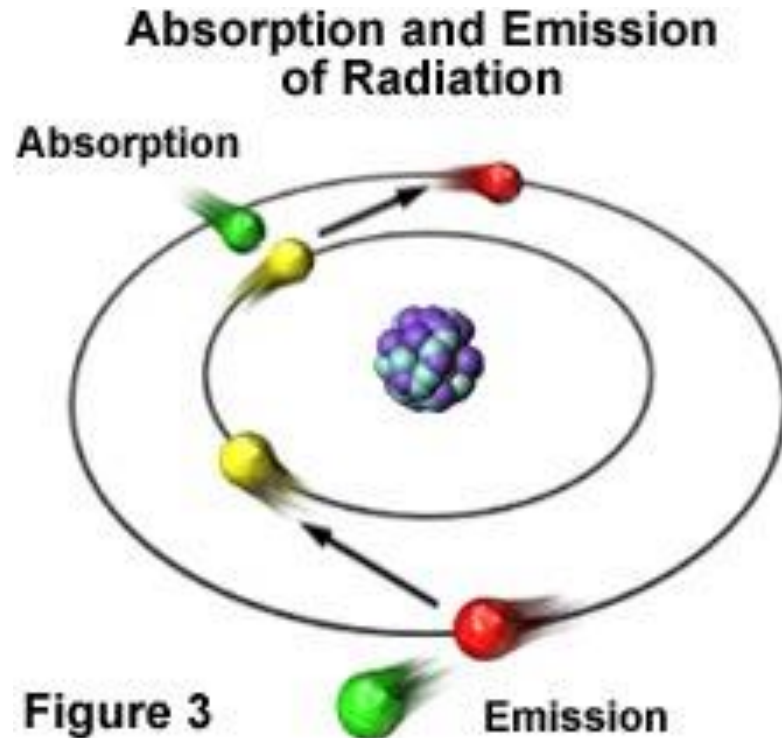
$$\Delta E = h\nu$$



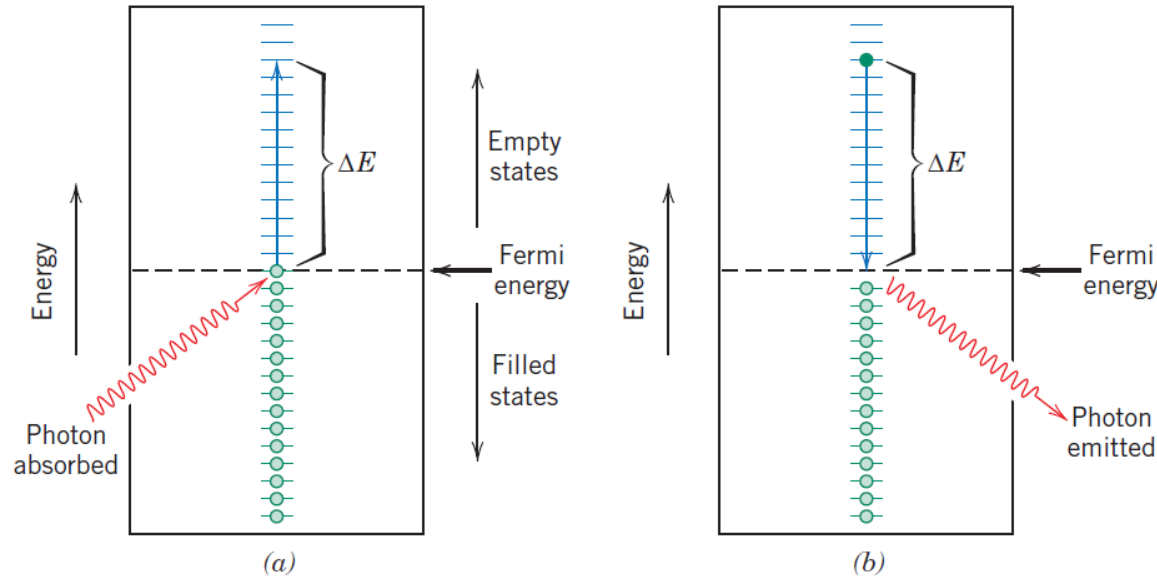
- Only photons of frequencies corresponding to the possible ΔE 's for the atom can be absorbed by electron transitions,
- All of a photon's energy is absorbed in each excitation event

- A second important concept is that a stimulated electron cannot remain in an excited state,
- After a short time, it falls or decays back into its ground state, or unexcited level, with a **reemission** of electromagnetic radiation,
- Several decay paths are possible. In any case, there must be a conservation of energy for absorption and emission electron transitions.

The optical characteristics of solid materials that relate to **absorption** and **emission** of electromagnetic radiation are explained in terms of the electron band structure of the material and the principles relating to electron transitions.



Optical Properties of Metals: Absorption



- In both cases a high-energy band is only partially filled with electrons.
- **Metals are opaque because** the incident radiation having frequencies within the visible range excites electrons into unoccupied energy states above the Fermi energy.
- As a consequence, the incident radiation is absorbed,
- Total absorption is within a very thin outer layer, usually less than 0.1μ thus only metallic films thinner than 0.1μ are capable of transmitting visible light.

Light Absorption

The amount of light absorbed by a material is calculated using Beer's Law

$$I'_T = I'_0 e^{-\beta \ell}$$

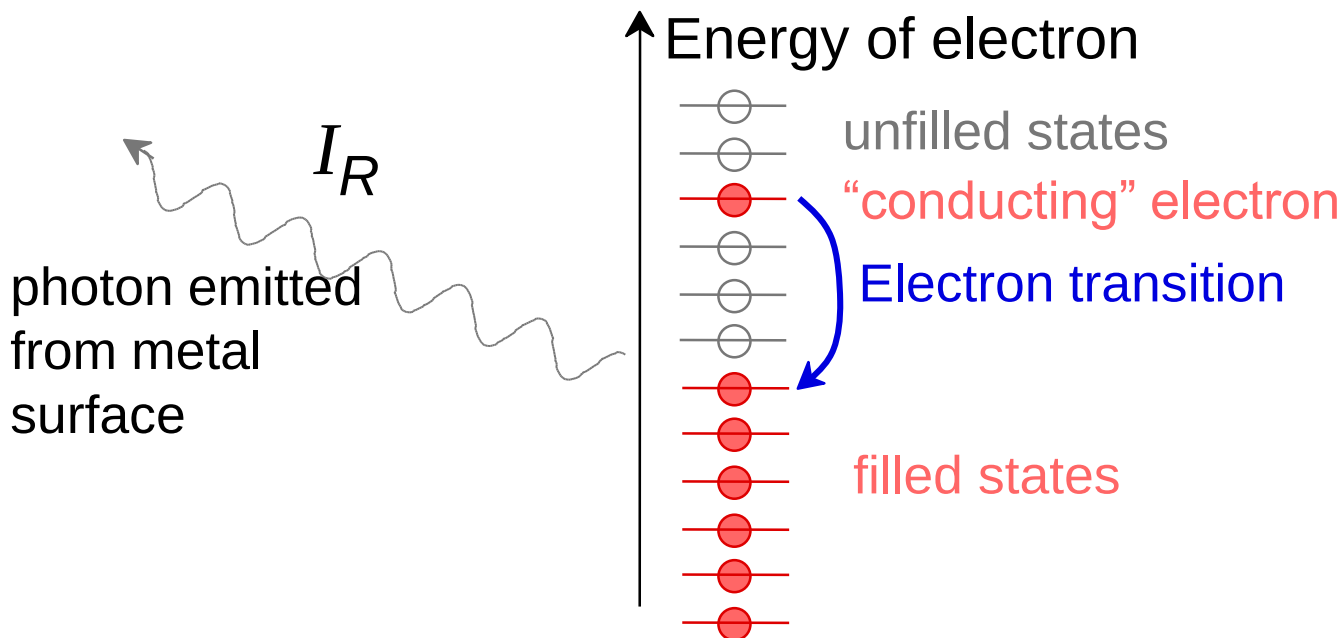
β = absorption coefficient, cm^{-1}
 ℓ = sample thickness, cm
 I'_0 = incident light intensity
 I'_T = transmitted light intensity

Rearranging and taking the natural log of both sides of the equation leads to

$$\ln \left[\frac{I'_T}{I'_0} \right] = -\beta \ell$$

Reflection of Light for Metals

- Most of the absorbed radiation is reemitted from the surface in the form of visible light of the same wavelength, which appears as reflected light.
- **Electron transition** from an excited state produces a photon.



Reflection of Light for Metals (cont.)

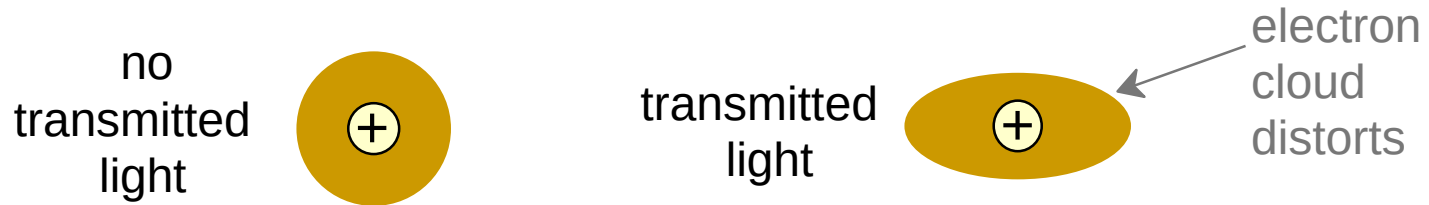
- **Reflectivity** = I_R/I_0 is between 0.90 and 0.95.
- Metal surfaces appear shiny
- Most of absorbed light is reflected at the same wavelength
- Small fraction of light may be absorbed
- Color of reflected light depends on wavelength distribution
 - Example: The metals copper and gold absorb light in blue and green => reflected light has gold color

Optical Properties of Nonmetals:

- By virtue of their electron energy band structures, **nonmetallic materials may be transparent to visible light.**
- Therefore, in addition to “**reflection**” and “**absorption**”, “**refraction**” and “**transmission**” phenomena also need to be considered.

Refraction

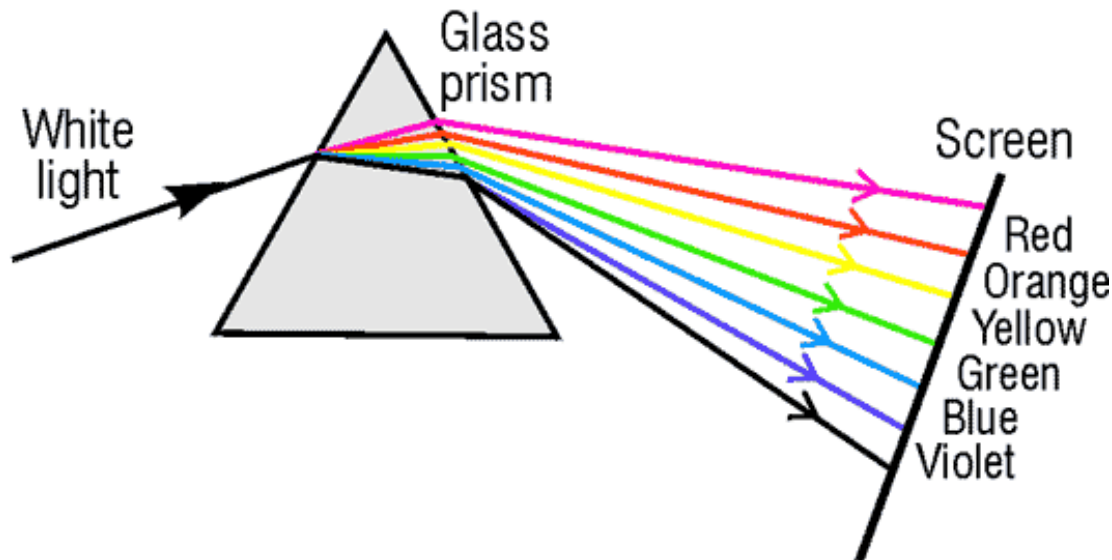
- Transmitted light distorts electron clouds.



- The velocity of light in a material is lower than in a vacuum (Decrease in velocity).

$$n = \text{index of refraction} \equiv \frac{c \text{ (velocity of light in vacuum)}}{v \text{ (velocity of light in medium)}}$$

- The magnitude of “ n ” (or the degree of bending) will depend on the wavelength of the light.
- This effect is graphically demonstrated by dispersion or separation of a beam of white light into its component colors by a glass prism.



- Each color is deflected by a different amount as it passes through the glass, which results in the separation of the colors.
- Not only does the index of **refraction** affect the optical path of light, but also it influences the fraction of incident light that is **reflected** at the surface.

- For transparent materials, there is a relation between the index of refraction and the dielectric constant:

$$n = \frac{c}{v} = \frac{\sqrt{\epsilon\mu}}{\sqrt{\epsilon_0\mu_0}} = \sqrt{\epsilon_r\mu_r}$$

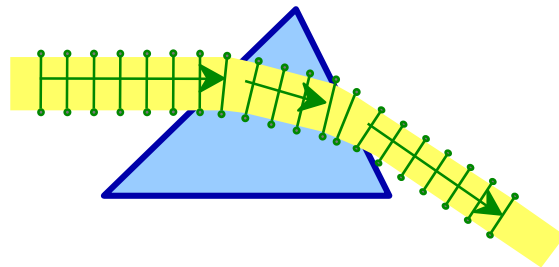
Dielectric Constant Relative Magnetic Permeability

most substances are only slightly magnetic: $\mu_r=1$

$$n \cong \sqrt{\epsilon_r}$$

- Since the retardation of electromagnetic radiation in a medium results from electronic polarization, the size of the constituent atoms or ions has a considerable influence on the magnitude of this effect.
- The larger an atom or ion, the greater will be the electronic polarization, the slower the velocity, and the greater the index of refraction.
- For **crystalline ceramics that have cubic crystal structures**, and for glasses, the index of refraction is independent of crystallographic direction.
- For **noncubic crystals**, on the other hand, have an anisotropic n ; the index is greatest along the directions that have the highest density of ions.

- Adding large ions (e.g., **lead**) to glass decreases the speed of light in the glass.
- Light can be “bent” as it passes through a transparent prism



Material	n
Typical glasses ca.	1.5 -1.7
Plastics	1.3 -1.6
PbO (Litharge)	2.67
Diamond	2.41

Reflection

- When light radiation passes from one medium into another having a different index of refraction, some of the light is scattered at the interface between the two media even if both are transparent.
- The **reflectivity** R represents the fraction of the incident light that is reflected at the interface.

$$R = \frac{I_R}{I_0}$$

Intensity of reflected beam

Intensity of Incident beam

$$R = \left(\frac{n_2 - n_1}{n_2 + n_1} \right)^2$$

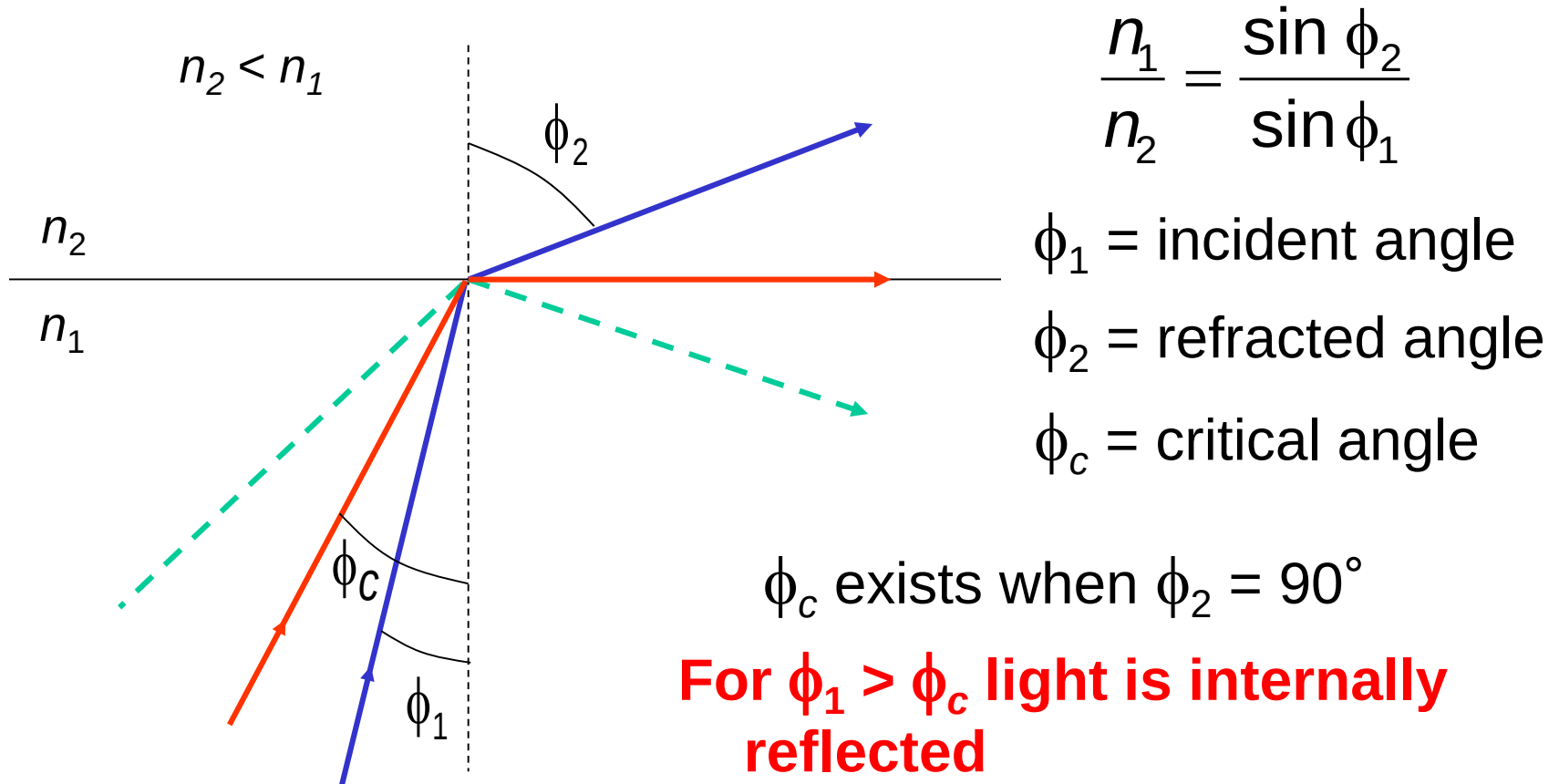
Reflection

- When light is transmitted from a vacuum or air into a solid s ,

$$R = \left(\frac{n_s - 1}{n_s + 1} \right)^2$$

- The higher the index of refraction of the solid, the greater is the reflectivity.
- Just as the “**index of refraction**” of a solid depends on the wavelength of the incident light, so also does the “**reflectivity**” vary with wavelength.

Total Internal Reflectance

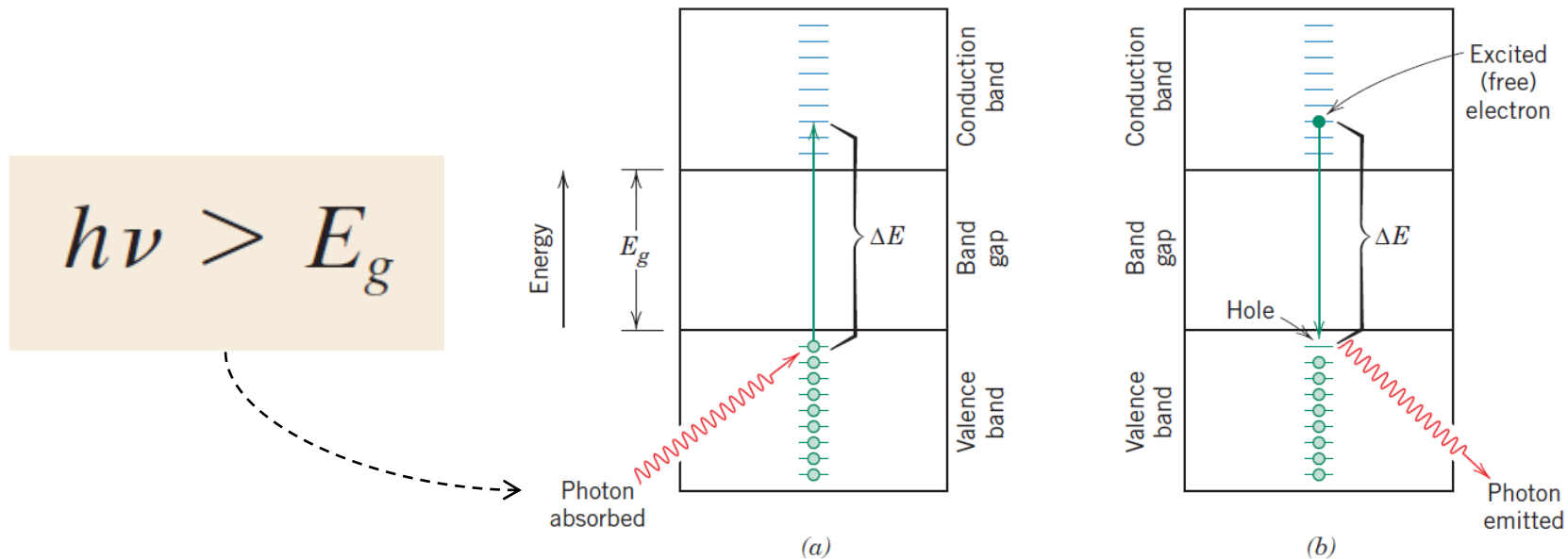


- **Fiber optic cables** are clad in low n material so that light will experience total internal reflectance and not escape from the optical fiber.

Absorption

- In principle, light radiation is absorbed in **nonmetallic materials** by two basic mechanisms, which also influence the transmission characteristics of these nonmetals:
- **One of these is electronic polarization.** Absorption by electronic polarization is important only at light frequencies in the vicinity of the relaxation frequency of the constituent atoms.
- **The other mechanism involves valence band-conduction band electron transitions,** which depend on the electron energy band structure of the material: Band structures for semiconductors and insulators.

- Absorption of a photon of light may occur by the promotion or excitation of an electron from the nearly filled valence band, across the band gap, and into an empty state within the conduction band.

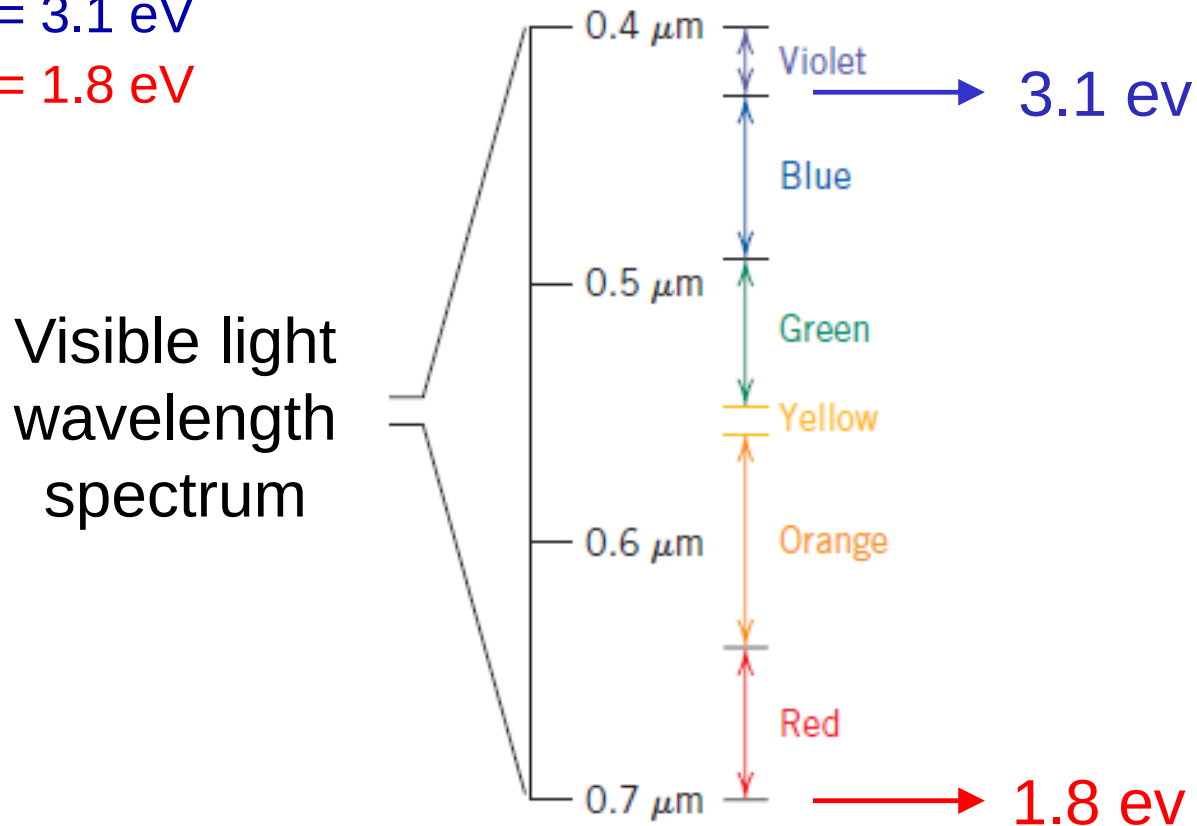


- These excitations with the accompanying absorption can take place only if the photon energy is greater than that of the band gap.

•All visible light is absorbed by “valence band to conduction band electron transitions” for those **semiconducting materials** that have band gap energies less than about **1.8 eV**; thus, these materials are **opaque**.

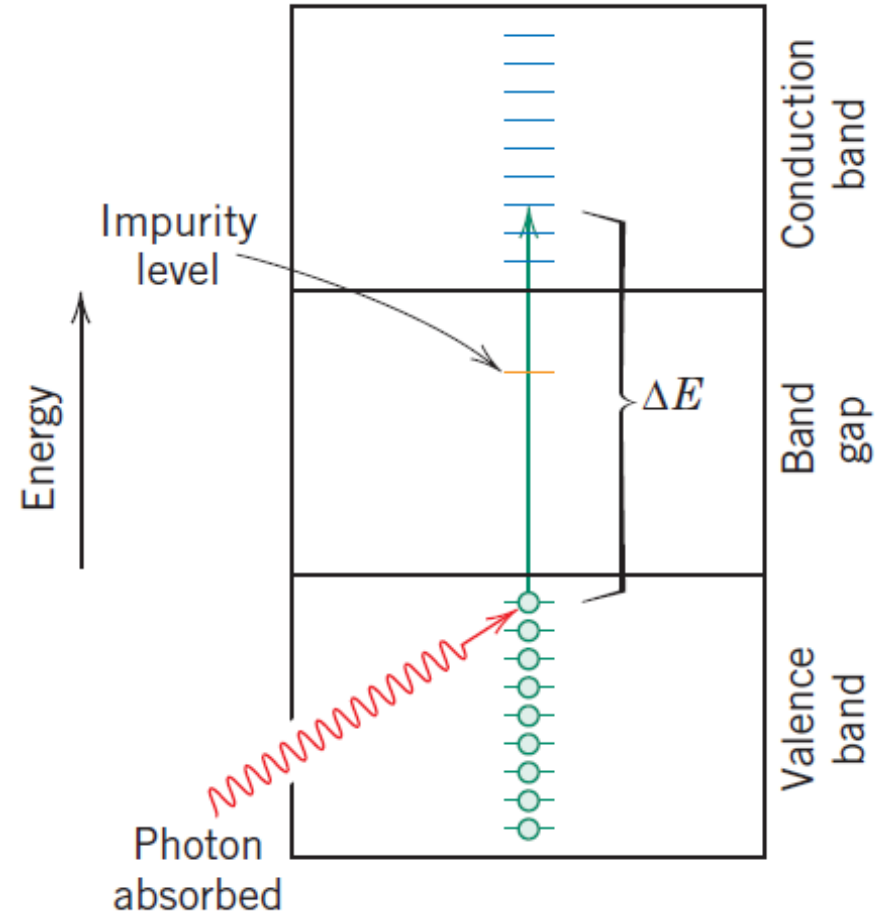
blue light: $h\nu = 3.1 \text{ eV}$

red light: $h\nu = 1.8 \text{ eV}$

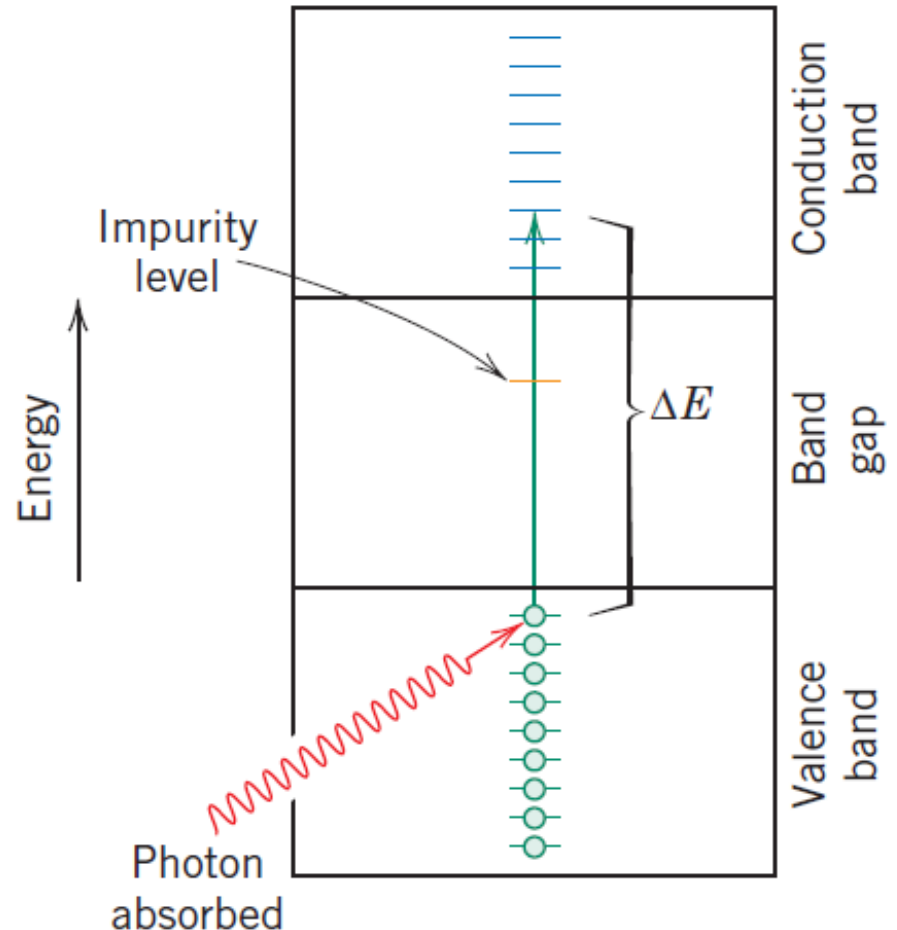


- Interactions with light radiation can also occur in **dielectric solids** having wide band gaps, involving other than valence band-conduction band electron transitions.

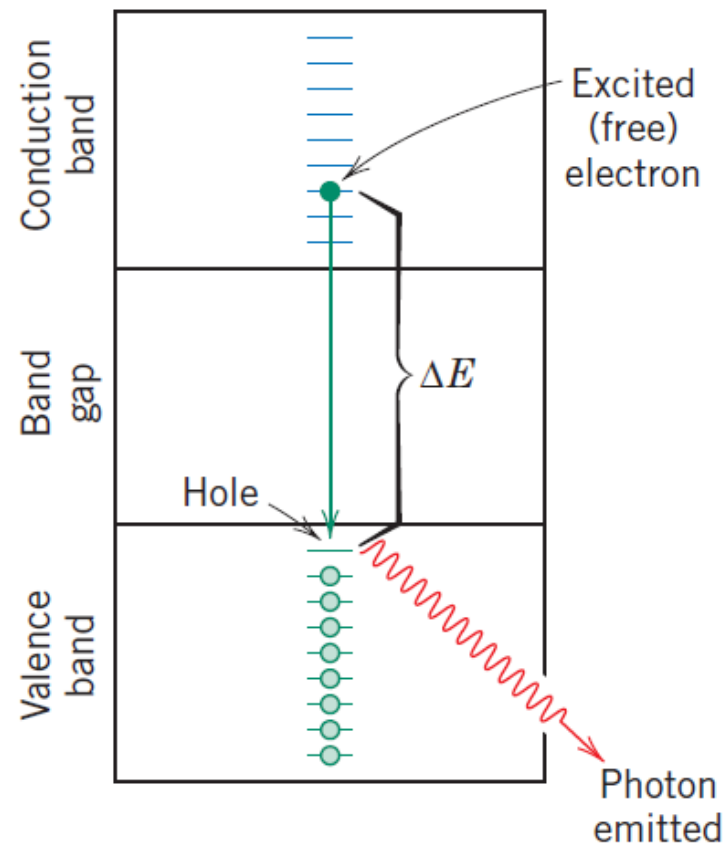
- If impurities or other electrically active defects are present, electron levels within the band gap may be introduced (such as the donor and acceptor levels, **except that they lie closer to the center of the band gap**).



- Light radiation of specific wavelengths may be emitted as a result of electron transitions involving these levels within the band gap.
- For example, according to figure, which shows the valence band - conduction band electron excitation for a material that has one such impurity level.
- The electromagnetic energy that was absorbed by this electron excitation must be dissipated in some manner; several mechanisms are possible:

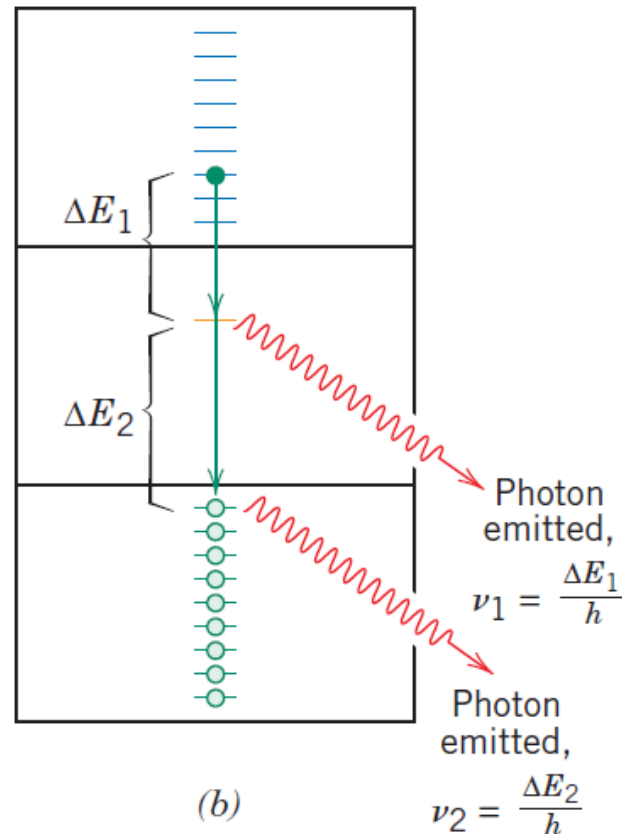


(1): Dissipation may occur via direct electron and hole recombination according to the reaction:

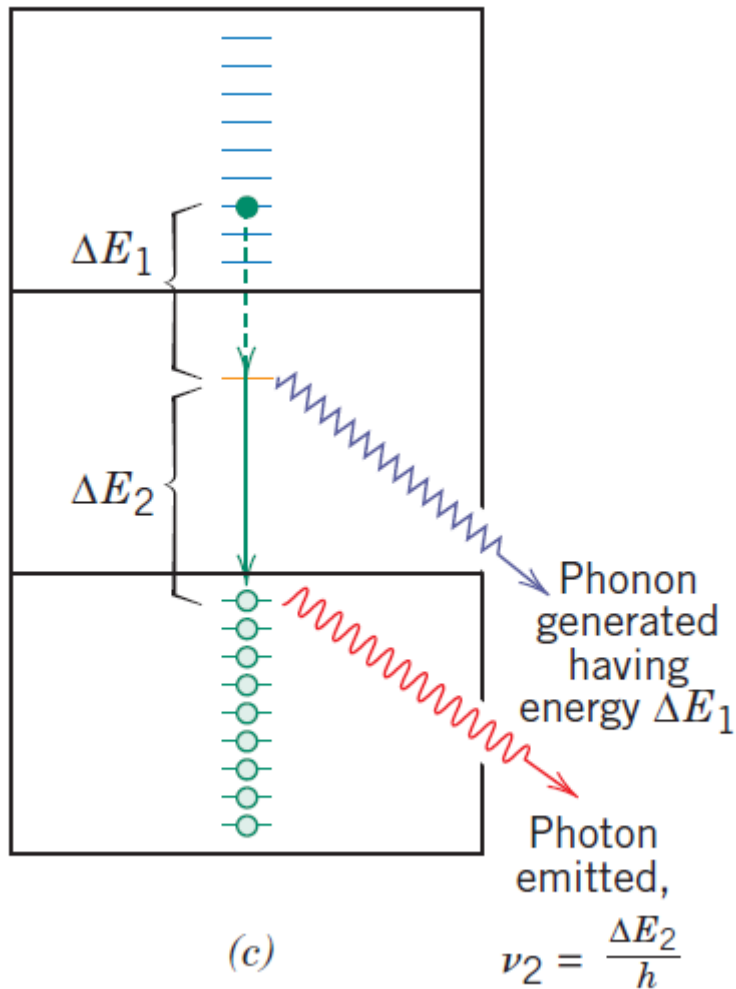


(2-1): Multiple-step electron transitions:

- The emission of two photons;
 - one is emitted as the electron drops from a state in the conduction band to the impurity level,
 - the other as it decays back into the valence band.



(2-2): Multiple-step electron transitions:



- One of the transitions may involve the generation of a **phonon**, wherein the associated energy is dissipated in the form of “**heat**”.

Selected Light Absorption in Semiconductors

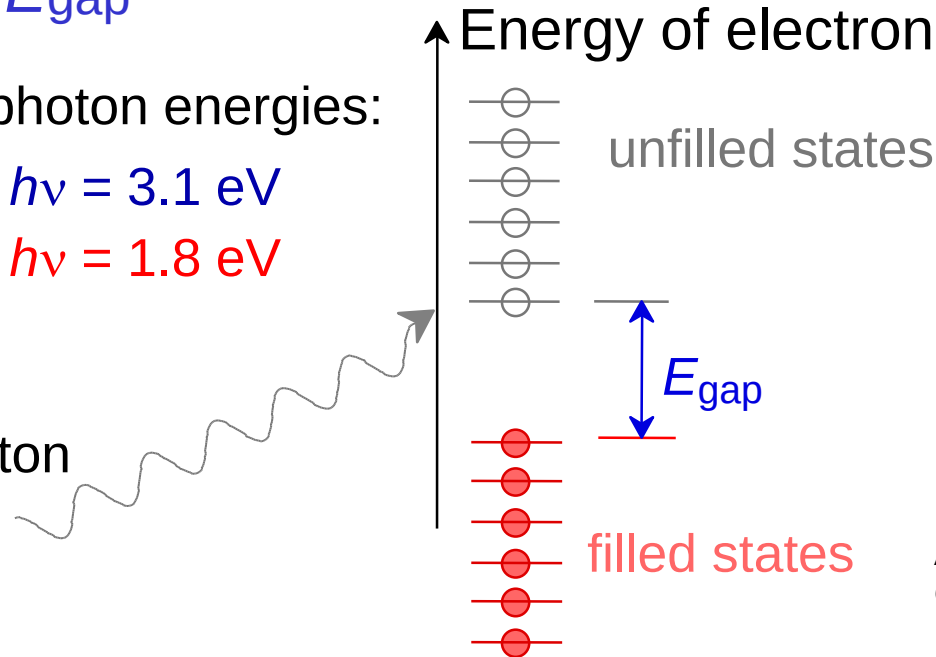
Absorption of light of frequency ν by electron transition occurs if $h\nu > E_{\text{gap}}$

Examples of photon energies:

blue light: $h\nu = 3.1 \text{ eV}$

red light: $h\nu = 1.8 \text{ eV}$

incident photon
energy $h\nu$



Adapted from Fig. 21.5(a),
Callister & Rethwisch 8e.

- If $E_{\text{gap}} < 1.8 \text{ eV}$, all light absorbed; material is **opaque** (e.g., Si, GaAs)
- If $E_{\text{gap}} > 3.1 \text{ eV}$, no light absorption; material is **transparent** and **colorless** (e.g., diamond)
- If $1.8 \text{ eV} < E_{\text{gap}} < 3.1 \text{ eV}$, partial light absorption; material is **colored**

Transmission:

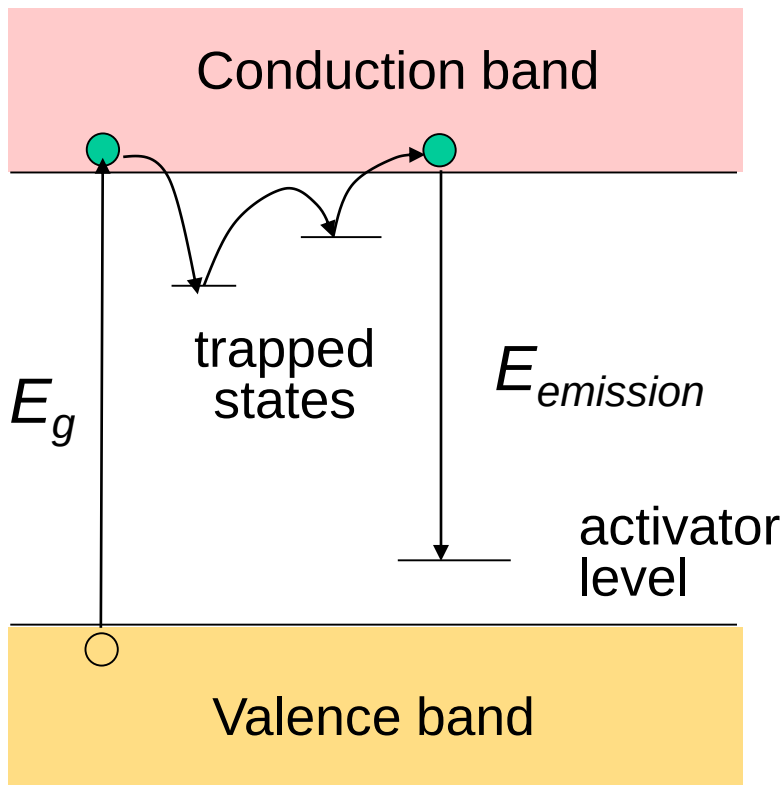
- The phenomena of absorption, reflection, and transmission may be applied to the passage of light through a transparent solid.
- The fraction of incident light that is transmitted through a transparent material depends on the losses that are incurred by absorption and reflection.



I_0 : Intensity of beam
 l : Thickness
 R : Reflection
 β : Absorption coefficient

Luminescence

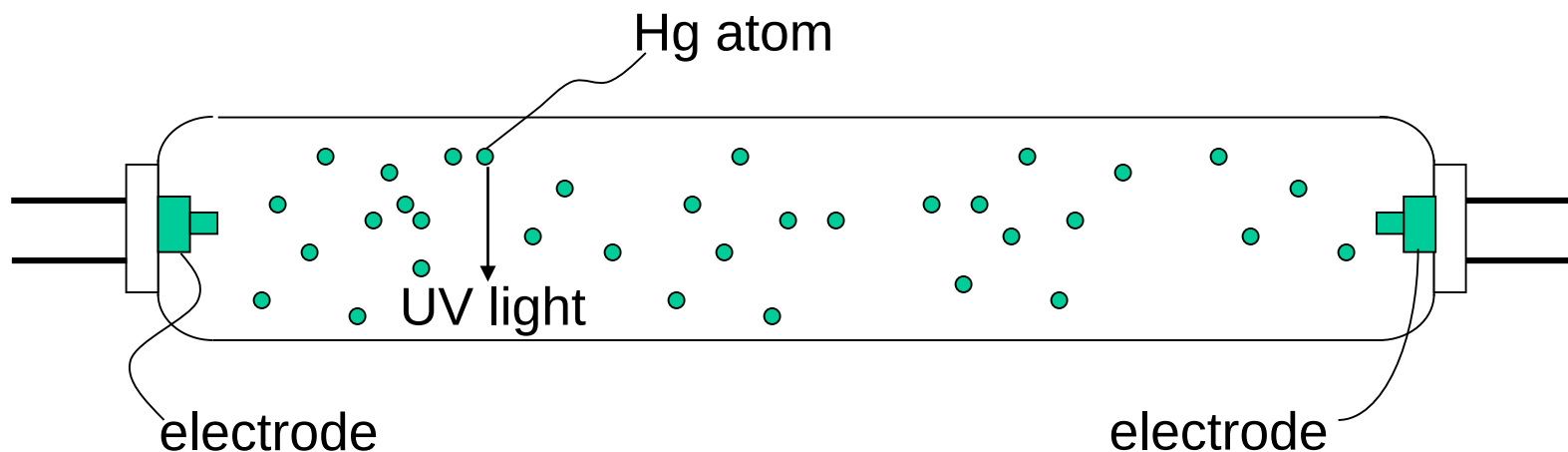
- **Luminescence** – reemission of light by a material
 - Material absorbs light at one frequency and reemits it at another (lower) frequency.
 - Trapped (donor/acceptor) states introduced by impurities/defects



- If residence time in trapped state is relatively long ($> 10^{-8}$ s)
-- **phosphorescence**
- For short residence times ($< 10^{-8}$ s)
-- **fluorescence**

Example: Toys that glow in the dark. Charge toys by exposing them to light. Reemission of light over time: **phosphorescence**

Photoluminescence



- Arc between electrodes excites electrons in mercury atoms in the lamp to higher energy levels.
- As electron falls back into their ground states, UV light is emitted (e.g., suntan lamp).
- Inside surface of tube lined with material that absorbs UV and reemits visible light
 - For example, $\text{Ca}_{10}\text{F}_2\text{P}_6\text{O}_{24}$ with 20% of F^- replaced by Cl^-

The LASER

- The laser generates light waves that are in phase (coherent) and that travel parallel to one another

– LASER

- Light
 - Amplification by
 - Stimulated
 - Emission of
 - Radiation
- Operation of laser involves a population inversion of energy states process

- All the radiative electron transitions heretofore discussed are **spontaneous**; that is an electron falls from a high energy state to a lower one without any external provocation.
- These transition events occur independently of one another and at random times, producing radiation that is incoherent (*uyumsuz*); that is, the light waves are out of phase with one another.
- With laser's, however, coherent (*uyumlu*) light is generated by electron transitions initiated by an **external stimulus**.

Light **A**mplification by **S**timulated **E**mission of **R**adiation

Other Applications - Optical Fibers

Schematic diagram showing components of a fiber optic communications system

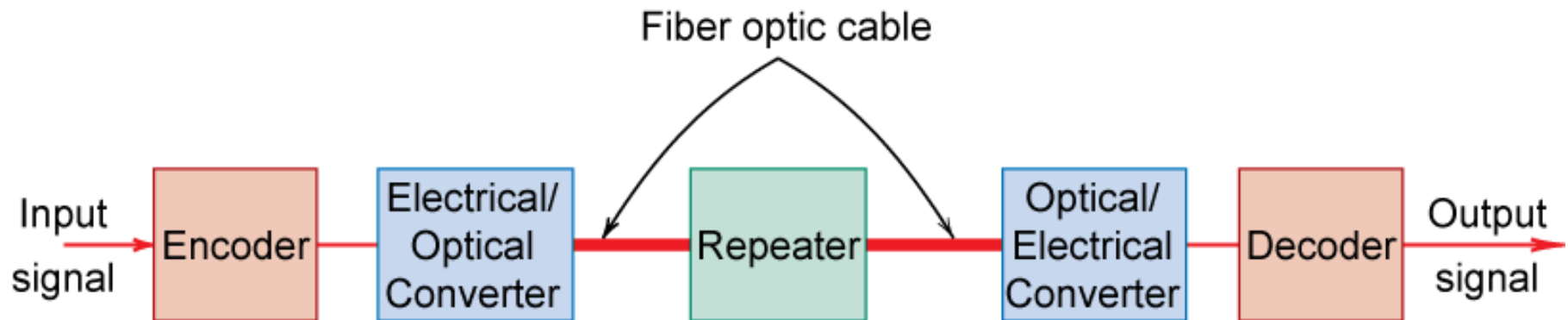
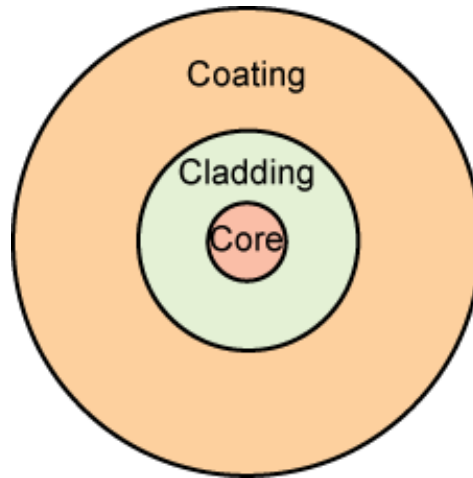


Fig. 21.18, *Callister & Rethwisch 8e.*

Optical Fibers (cont.)

- fibers have diameters of 125 μm or less
- plastic cladding 60 μm thick is applied to fibers



- **Fiber optic cables** are clad in low n material so that light will experience total internal reflectance and not escape from the optical fiber.

Optical Fiber Designs

Step-index Optical Fiber

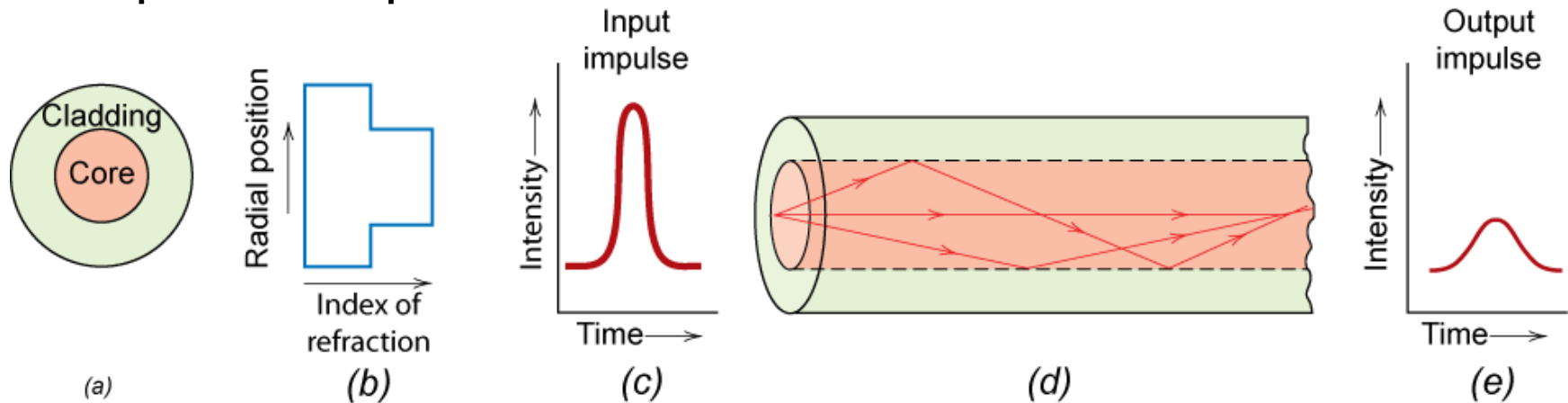


Fig. 21.21, Callister & Rethwisch 8e.

Graded-index Optical Fiber

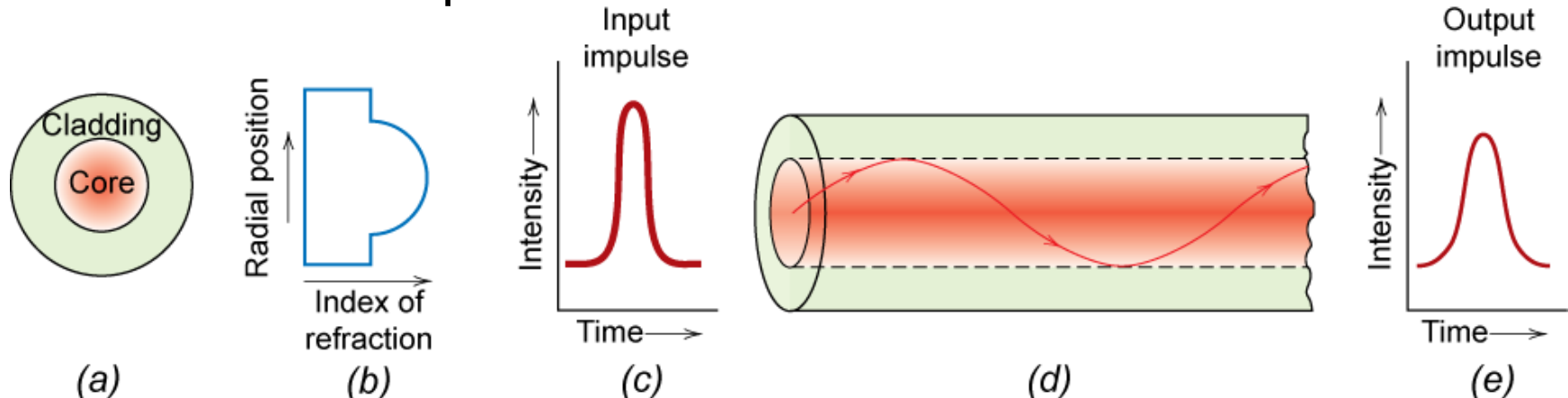


Fig. 21.22, Callister & Rethwisch 8e.

SUMMARY

- Light radiation impinging on a material may be **reflected** from, **absorbed** within, and/or **transmitted** through
- Light transmission characteristics:
 - **transparent**, **translucent**, **opaque**
- Optical properties of **metals**:
 - opaque and highly reflective due to electron energy band structure.
- Optical properties of **non-Metals**:
 - for $E_{gap} < 1.8$ eV, absorption of all wavelengths of light radiation
 - for $E_{gap} > 3.1$ eV, no absorption of visible light radiation
 - for $1.8 \text{ eV} < E_{gap} < 3.1$ eV, absorption of some range of light radiation wavelengths
 - color determined by wavelength distribution of transmitted light
- Other important optical applications/devices:
 - luminescence, **photoconductivity**, **light-emitting diodes**, **solar cells**, lasers, and optical fibers